Problem Set 1: Euclid and the Architecture of the Integers Worked Solutions

Divisibility, gcds, primes, and unique factorization

80 points + 10 optional challenge points

Purpose. This assignment develops the two complementary habits of elementary number theory: compute examples efficiently, then explain why the method works. Show the divisions in every use of Euclid's algorithm and name the theorem used in every proof.
You may use: the division algorithm, the Euclidean algorithm, Bézout's identity, Euclid's lemma, and the fundamental theorem of arithmetic. Unless a problem explicitly asks for a decimal answer, exact arithmetic is expected.

Instructor note

Problem 3 is deliberately removable. If Bézout's identity has not yet been covered, assign 3(a) as a guided extension and postpone 3(b)–(c). The remaining problems do not depend on Problem 3.

I. Divisibility and Euclid's algorithm

1. The invariant behind the algorithm. [6 points] Let $a, b, d, m, n \in \mathbb{Z}$.

(a) Prove that if $d \mid a$ and $d \mid b$, then $d \mid (ma + nb)$.

(b) Suppose $a = qb + r$. Prove that (a, b) and (b, r) have exactly the same common divisors. Conclude that $\gcd(a, b) = \gcd(b, r)$.

Solution

(a) Since $d \mid a$ and $d \mid b$, write $a = du$ and $b = dv$ for some $u, v \in \mathbb{Z}$. Then

$$ma + nb = mdu + ndv = d(mu + nv),$$

so $d \mid (ma + nb)$.

(b) If d divides both a and b, part (a), with coefficients 1 and $-q$, shows that $d \mid a - qb = r$. Thus d divides both b and r. Conversely, if $d \mid b$ and $d \mid r$, then $d \mid qb + r = a$. The pairs therefore have the same common divisors, so their greatest positive common divisors are equal.

2. Three runs of Euclid's algorithm. [9 points] Compute each gcd, displaying every division. For part (c), describe the pattern in the remainders.

(a) $\gcd(2024, 748)$

(b) $\gcd(12345, 6789)$

(c) $\gcd(4181, 2584)$

Solution

(a)

$$\begin{aligned} 2024 &= 2 \cdot 748 + 528, \\ 528 &= 2 \cdot 220 + 88, \\ 88 &= 2 \cdot 44. \end{aligned} \qquad \begin{aligned} 748 &= 1 \cdot 528 + 220, \\ 220 &= 2 \cdot 88 + 44, \end{aligned}$$

Hence $\gcd(2024, 748) = 44$.

(b)

$$\begin{aligned} 12345 &= 1 \cdot 6789 + 5556, \\ 5556 &= 4 \cdot 1233 + 624, \\ 624 &= 1 \cdot 609 + 15, \\ 15 &= 1 \cdot 9 + 6, \\ 6 &= 2 \cdot 3. \end{aligned} \qquad \begin{aligned} 6789 &= 1 \cdot 5556 + 1233, \\ 1233 &= 1 \cdot 624 + 609, \\ 609 &= 40 \cdot 15 + 9, \\ 9 &= 1 \cdot 6 + 3, \end{aligned}$$

Thus $\gcd(12345, 6789) = 3$.

(c)

$$\begin{aligned} 4181 &= 1 \cdot 2584 + 1597, \\ 1597 &= 1 \cdot 987 + 610, \\ 610 &= 1 \cdot 377 + 233, \\ 233 &= 1 \cdot 144 + 89, \\ 89 &= 1 \cdot 55 + 34, \\ 34 &= 1 \cdot 21 + 13, \\ 13 &= 1 \cdot 8 + 5, \\ 5 &= 1 \cdot 3 + 2, \\ 2 &= 2 \cdot 1. \end{aligned} \qquad \begin{aligned} 2584 &= 1 \cdot 1597 + 987, \\ 987 &= 1 \cdot 610 + 377, \\ 377 &= 1 \cdot 233 + 144, \\ 144 &= 1 \cdot 89 + 55, \\ 55 &= 1 \cdot 34 + 21, \\ 21 &= 1 \cdot 13 + 8, \\ 8 &= 1 \cdot 5 + 3, \\ 3 &= 1 \cdot 2 + 1, \end{aligned}$$

Therefore $\gcd(4181, 2584) = 1$. The remainders run backward through the Fibonacci numbers; each quotient before the last is 1.

3. Running Euclid backward. [12 points]

(a) Use the divisions from Problem 2(a) to express 44 as $2024x + 748y$ with $x, y \in \mathbb{Z}$.

(b) Find one integer solution of $2024x + 748y = 132$, and then find all integer solutions.

(c) Explain why $2024x + 748y = 100$ has no integer solution.

Solution

(a) Reverse the divisions:

$$\begin{aligned} 44 &= 220 - 2 \cdot 88 \\ &= 220 - 2(528 - 2 \cdot 220) = 5 \cdot 220 - 2 \cdot 528 \\ &= 5(748 - 528) - 2 \cdot 528 = 5 \cdot 748 - 7 \cdot 528 \\ &= 5 \cdot 748 - 7(2024 - 2 \cdot 748) = -7 \cdot 2024 + 19 \cdot 748. \end{aligned}$$

Thus $(x, y) = (-7, 19)$.

(b) Multiplying the identity in (a) by 3 gives one solution $(x_0, y_0) = (-21, 57)$. Since $2024/44 = 46$ and

Solution (continued)

$748/44 = 17$, all solutions are

$$x = -21 + 17t, \qquad y = 57 - 46t, \qquad t \in \mathbb{Z}.$$

Indeed, the changes contribute $2024(17) - 748(46) = 0$, and every solution differs from the displayed one by such a change.

(c) Every integer linear combination of 2024 and 748 is divisible by their gcd, 44. Since $44 \nmid 100$, no solution exists.

4. Useful gcd identities. [8 points]

(a) Prove that $\gcd(a + kb, b) = \gcd(a, b)$ for every $k \in \mathbb{Z}$.

(b) Use part (a), without factoring, to prove that $\gcd(2n + 1, n) = 1$ for every positive integer n.

(c) Prove that two consecutive positive integers are relatively prime.

Solution

(a) A common divisor of a and b divides $(a + kb)$ and b. Conversely, a common divisor of $a + kb$ and b divides $(a + kb) - kb = a$. The pairs have exactly the same common divisors.

(b) Apply (a) with $a = 1$, $b = n$, and $k = 2$: $\gcd(2n + 1, n) = \gcd(1, n) = 1$.

(c) A common divisor of n and $n + 1$ divides their difference 1. Hence their gcd is 1.

5. Why the algorithm stops and succeeds. [7 points] In a paragraph, prove the Euclidean algorithm: explain both why repeated division must terminate and why the last nonzero remainder is the gcd of the original two positive integers.

Solution

At each step the new remainder is a nonnegative integer strictly smaller than the preceding positive remainder. There cannot be an infinite strictly decreasing sequence of nonnegative integers, so eventually a remainder is zero. By Problem 1(b), replacing a pair by the divisor and remainder does not change its common divisors. Thus every pair in the algorithm has the same gcd as the original pair. At the final nonzero step we have $r_{k-1} = qr_k$, so the common divisors of r_{k-1} and r_k are precisely the divisors of r_k. Their greatest positive common divisor is therefore r_k, the last nonzero remainder.

II. Primes and unique factorization

6. Prime-exponent coordinates. [12 points]

(a) Find the prime factorizations of 7560 and 8820.

(b) Use the exponent vectors, without applying Euclid's algorithm, to find their gcd and lcm.

(c) How many positive divisors does 7560 have? Justify the counting rule you use.

(d) Verify numerically that $\gcd(a, b)\,\mathrm{lcm}(a, b) = ab$ for this pair.

Solution

(a)
$$7560 = 2^3 \cdot 3^3 \cdot 5 \cdot 7, \qquad 8820 = 2^2 \cdot 3^2 \cdot 5 \cdot 7^2.$$

(b) Coordinatewise minima and maxima give
$$\gcd(7560, 8820) = 2^2 \cdot 3^2 \cdot 5 \cdot 7 = 1260$$
and
$$\operatorname{lcm}(7560, 8820) = 2^3 \cdot 3^3 \cdot 5 \cdot 7^2 = 52920.$$

(c) A positive divisor has the form $2^a 3^b 5^c 7^d$ with $0 \le a \le 3$, $0 \le b \le 3$, and $0 \le c, d \le 1$. The choices are independent, so there are $(3+1)(3+1)(1+1)(1+1) = 64$ positive divisors.

(d)
$$1260 \cdot 52920 = 66{,}679{,}200 = 7560 \cdot 8820.$$

7. Euclid's lemma, the hinge of uniqueness. [8 points]

(a) Let p be prime. Prove that if $p \mid ab$, then $p \mid a$ or $p \mid b$. You may use Bézout's identity.

(b) Deduce that if $p \mid a^m$ for an integer $m \ge 1$, then $p \mid a$.

(c) In two or three sentences, explain where part (a) enters the uniqueness proof in the fundamental theorem of arithmetic.

Solution

(a) If $p \mid a$, there is nothing to prove. Otherwise $\gcd(p, a) = 1$, since the only positive divisors of p are 1 and p. Bézout gives $px + ay = 1$. Multiplying by b yields $pbx + aby = b$. Both terms on the left are divisible by p (the second because $p \mid ab$), so $p \mid b$.

(b) Apply part (a) to $a^m = a \cdot a^{m-1}$. Either $p \mid a$, or $p \mid a^{m-1}$. Repeating this argument eventually gives $p \mid a$.

(c) If $p_1 \cdots p_r = q_1 \cdots q_s$, then p_1 divides the product on the right. Euclid's lemma implies that p_1 equals one of the prime factors q_j. Cancel that common factor and repeat; this forces the two lists of primes to agree up to order.

8. Two consequences of prime factorization. [8 points]

(a) Suppose $\gcd(a, b) = 1$ and ab is a perfect square. Prove that both a and b are perfect squares.

(b) Prove that $\sqrt{p}$ is irrational for every prime p.

Solution

(a) In the prime factorizations of a and b, no prime occurs in both, because $\gcd(a, b) = 1$. Since ab is a square, every prime has even exponent in the factorization of the product. Each exponent comes entirely from either a or b, so every exponent in each factor is even. Hence a and b are squares.

(b) Suppose $\sqrt{p} = m/n$ with positive integers m, n chosen relatively prime. Then $m^2 = pn^2$, so $p \mid m^2$. Problem 7(b) gives $p \mid m$; write $m = pk$. Substitution gives $p^2k^2 = pn^2$, hence $n^2 = pk^2$ and $p \mid n^2$. Again $p \mid n$, contradicting $\gcd(m, n) = 1$. Therefore $\sqrt{p}$ is irrational.

9. There is no last prime. [10 points]

(a) Give Euclid's proof that there are infinitely many primes. Your proof must explicitly explain why the constructed integer has a prime divisor and why that divisor is new.

(b) The first six primes are $2, 3, 5, 7, 11, 13$. Compute their product plus one and verify

$$2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1 = 30031 = 59 \cdot 509.$$

What common misunderstanding of Euclid's proof does this example correct?

Solution

(a) Suppose there were only finitely many primes, say $p_1, \ldots, p_k$. Form $N = p_1 p_2 \cdots p_k + 1$. Since $N > 1$, the existence part of the fundamental theorem of arithmetic says that N has a prime divisor q. But division of N by any p_i leaves remainder 1, so none of the listed primes divides N. Thus q is a prime missing from the supposedly complete list, a contradiction.

(b) The product is 30030, so adding 1 gives 30031. Direct multiplication verifies $59 \cdot 509 = 59(500 + 9) = 29500 + 531 = 30031$. This shows that the number constructed in Euclid's proof need not itself be prime. The proof needs only a prime divisor of that number which is absent from the proposed finite list.

Optional challenge

10. Infinitely many primes leaving remainder 3 upon division by 4. [10 extra points] Suppose, for contradiction, that the complete list of such primes is $p_1, \ldots, p_k$, and consider

$$N = 4p_1 p_2 \cdots p_k - 1.$$

Prove that some prime divisor of N leaves remainder 3 upon division by 4, and that this divisor is not one of the p_i. Conclude that there are infinitely many primes of the form $4n + 3$.

Solution

The integer N is odd and leaves remainder 3 upon division by 4. Every odd prime leaves remainder 1 or 3 upon division by 4. If every prime factor of N left remainder 1, then their product (with multiplicity) would also leave remainder 1, contradicting the fact that N leaves remainder 3. Thus at least one prime divisor q of N leaves remainder 3. For each p_i, division of N by p_i leaves remainder -1, so $p_i \nmid N$. Therefore q is a new prime of the form $4n + 3$, contradicting completeness of the list.

The arithmetic is the evidence; the proof is the explanation.